# Routing Protocol Configuration -- Guided Notes ANSWER KEY

Unit 2.4.1 | CompTIA Network+ N10-009 Obj. 2.4 | N10-008 2.4

For instructor use / distribute after students complete the notes.

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### Question 1

A _______ route is manually configured by an administrator and does not update automatically when the network changes. A _______ route is learned automatically through a routing protocol. The _______ route is used when no more specific match exists in the routing table.

*Answer: Static route; dynamic route; default route (0.0.0.0/0).*

### Question 2

Distance-vector routing protocols share their _______ table with directly connected neighbors at regular intervals. The two classic distance-vector protocols are _______ (used in small networks) and _______ (used in large enterprise networks).

*Answer: Routing table; RIP (Routing Information Protocol); EIGRP (Enhanced Interior Gateway Routing Protocol).*

### Question 3

OSPF is a _______ -state routing protocol. Instead of sharing routing tables, routers share _______ advertisements (LSAs) to build a complete map of the network called the _______. Each router then independently calculates the best path using Dijkstra's algorithm.

*Answer: Link-state; Link State Advertisements; LSDB (Link State Database).*

### Question 4

Administrative distance (AD) is used when a router learns about the same network from _______ sources. The route with the _______ AD value is preferred. List the AD values for: directly connected (___), static route (___), OSPF (___), RIP (___).

*Answer: Multiple (different) routing sources; lowest AD; Connected = 0; Static = 1; OSPF = 110; RIP = 120.*

### Question 5

BGP (Border Gateway Protocol) is the routing protocol that runs the _______. It is classified as an _______ gateway protocol because it routes between different autonomous systems. Unlike OSPF or EIGRP, BGP makes routing decisions based primarily on _______.

*Answer: Internet (public internet backbone); Exterior (EGP); path attributes and policy (not just metric/cost).*

### Question 6

Route redistribution allows a router to take routes learned from one protocol (e.g., OSPF) and advertise them into another (e.g., EIGRP). Why might this be necessary, and what is one risk of misconfigured redistribution?

*Answer: Redistribution is needed when two different routing protocols must share route information -- for example, when two companies with different protocols merge networks. A misconfigured redistribution can cause routing loops, where a route learned from one protocol is redistributed back into the original protocol with a different metric.*

**Question 7**

In OSPF, routers within the same _______ exchange LSAs directly. Area 0 is called the _______ area and all other areas must connect to it. This hierarchical design reduces the size of the _______ on each router.

*Answer: Area; backbone; LSDB (link state database).*

**Question 8 -- Real World Application**

REAL WORLD: A network engineer notices that traffic between two sites is taking an unexpected path -- traveling through a slower WAN link instead of a faster fiber connection. Both paths are learned via OSPF. What OSPF value would the engineer adjust to prefer the faster link, and how does OSPF calculate this value by default?

*Answer: The engineer adjusts the OSPF cost on the interface. By default, OSPF calculates cost as reference bandwidth divided by interface bandwidth (default reference = 100 Mbps). A gigabit interface and a 100 Mbps interface both get cost 1 -- which is why the reference bandwidth is often manually set higher. To prefer the fiber link, lower its cost or raise the cost on the slower WAN interface.*